AR-Based Tactical Helmet

Group Members: Muhammad Daud, Mustaeen Ur Rahman, Yammaan Ghouri, Zawar Hussain

Advisor: *Dr. Hadeed Ahmad Sher*

Institution: Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

1. Problem Statement

Modern infantry operations suffer from a critical information gap. Critical data — including GPS coordinates, compass headings, and situational telemetry — is distributed across separate handheld devices. Every time a soldier looks down at a GPS module or radio, they break their line of sight with the surrounding environment for 1 to 2 seconds. In a live tactical scenario, this brief loss of awareness is enough to be fatal.

This problem can be understood through the OODA Loop (Observe, Orient, Decide, Act) — the decision cycle that governs soldier performance in combat. Current systems break this loop: a soldier must stop observing their environment in order to orient themselves on a map. The result is a slow, fragmented decision cycle that puts the soldier at a tactical disadvantage.

Additionally, operating vehicle-mounted or remote weapon systems forces the gunner to either expose themselves to enemy fire or use non-intuitive joystick controls that introduce both latency and cognitive load. The gunner must mentally translate visual target position into joystick commands — a step that slows engagement and reduces accuracy.

Current indigenous solutions lack a unified Human-Machine Interface (HMI) that integrates thermal vision, navigation, and telemetry into a single heads-up display. There is a strategic need for a cost-effective, locally developed Soldier System that fuses Command & Control (C2) data directly into the operator's Field of View (FOV), keeping the OODA loop continuous and unbroken.

2. Initial Plan — Hardware Stack

- **Display:** Reflector-based AR HUD using the ST7789 IPS TFT display module (1.3 or 1.54 inches) for full RGB color output over fast SPI communication. The display reflects off a semi-transparent plexiglass visor using the Pepper's Ghost optical technique, projecting HUD graphics directly into the operator's field of view without blocking natural vision.
- **Sensor Suite:** Raspberry Pi Camera Module 3 (NoIR) with dedicated IR LEDs for primary low-light and night-vision operations. MLX90640 thermal sensor or **FLIR Lepton** integrated as a modular extension for thermal imaging capability.
- **Rear Camera:** Sony IMX291 Starvis sensor providing a live rearview Picture-in-Picture (PiP) feed, giving the operator 360-degree situational awareness without head rotation.
- **Processing Unit:** Raspberry Pi 5 (8GB) as the primary onboard computer for HUD rendering, sensor fusion, and video processing. An ESP32 microcontroller handles extended turret control via low-latency UDP communication over Wi-Fi.

- **IMU:** Adafruit 9-DOF BNO055 (primary choice) or MPU-9250 9-axis IMU. The 9-axis configuration is essential — a 6-axis IMU can track pitch and roll relative to gravity but cannot determine true North (Yaw) without magnetometer input. The BNO055 magnetometer provides a fixed Earth-reference to correct yaw drift continuously.
- **Turret Extension Motors:** NEMA 17 Stepper Motors for smooth micro-stepping control, with TB6600 stepper motor drivers. MG966R high-torque servo motors as an alternative for pitch movements. A hybrid configuration can use NEMA 17 for yaw and MG966R for pitch.
- **Software:** Python as the primary development language. PyGame library for HUD rendering and 2D minimap display (fast rotation and real-time centering). OpenCV for video feed handling, rearview camera resizing, and night-vision processing. Kalman filtering applied to raw IMU data to eliminate sensor drift and stabilize heading output.
- **Power:** Helmet unit powered by a Li-Po battery bank. Turret unit powered by a separate 12V supply to handle high-torque motor current requirements.

3. Methodology

The system is designed as a modular platform to demonstrate sensor fusion, aligned with modern Optronics standards. It is structured across three subsystems:

Subsystem A: C2 & Optronics Interface (Core Project — The Helmet)

- **Augmented Reality HUD:** A Pepper's Ghost waveguide-style display overlays real-time telemetry — heading, pitch, yaw, compass bearing — and tactical minimap data directly into the operator's FOV via a transparent plexiglass visor. The HUD interface is designed in the style of modern military AR systems (comparable to Anduril EagleEye) with semi-transparent neon green elements that do not obstruct natural vision.
- **Rearview Camera PiP:** The Sony IMX291 rear camera feeds a live Picture-in-Picture video window into the HUD, providing continuous 360-degree awareness without requiring the operator to turn their head. The PiP window occupies a small corner of the FOV, functioning like a rearview mirror — large enough to be useful, small enough not to distract.
- **Thermal Vision Display:** A circular thermal display window in the HUD shows infrared heat signatures of people and objects ahead. This is a modular extension for night and low-visibility operations, activated independently of the primary RGB camera feed.
- **Multi-Vision Ready:** The architecture handles standard RGB and NoIR cameras for baseline operations, and is structured to accept thermal core data for future night-operation capabilities mimicking the Shibli Tarsier/Skua standard.

Subsystem B: Inertial Tracking & Telemetry (Future Addition)

- **9-Axis IMU Tracking:** The BNO055 IMU captures accelerometer (linear force/gravity), gyroscope (rotational velocity), and magnetometer (absolute heading) data simultaneously. This provides the exact gaze vector needed to drive the digital compass, map orientation, and turret aiming angle.
- **Kalman Filter Stabilization:** Raw IMU data is processed through a Kalman Filter algorithm that fuses the stable-but-slow accelerometer data with the fast-but-drifting gyroscope data to produce a clean, accurate heading estimate. This

eliminates sensor drift and human jitter, keeping HUD telemetry stable even during rapid head movements.

- **Zeroing Calibration:** A dedicated zeroing function allows the operator to press a single button while looking straight ahead to define the 0,0 reference coordinate. This calibrates the system to the individual wearer and resets accumulated drift at the start of each operation.

Subsystem C: Scalable Turret Integration (Extended Modular Feature)

- **Master-Slave Head-Tracking:** The helmet acts as the Master and the turret as the Slave in a head-tracking control system. The turret mirrors the operator's head orientation in real time, binding the weapon's line of sight to the operator's natural gaze. This eliminates the cognitive load of joystick translation and leverages the body's natural proprioception for faster, more accurate targeting.
- **PID Control Loop:** A PID (Proportional-Integral-Derivative) controller governs turret movement. The P term drives fast response proportional to heading error. The I term eliminates residual small errors that prevent the turret from reaching exact position. The D term prevents overshoot and oscillation, acting as a damper on inertia.
- **Secure UDP Datalink:** Low-latency wireless communication via UDP over Wi-Fi transmits processed aiming coordinates from the helmet to the ESP32 turret controller. For the prototype, this operates within lab range. A real deployment would use tactical radio waveforms or LoRa for long-range coordinate-only transmission.
- **Watchdog Safety:** A watchdog timer in the turret firmware detects signal loss. If no valid heading packet is received for 500ms, the turret automatically centers itself and locks to prevent uncontrolled movement.
- **Latency Target:** End-to-end system latency from head movement to turret response is targeted at under 100ms. Above this threshold, the operator perceives a disconnect between gaze and weapon position, degrading targeting performance and causing disorientation.

4. Expected Results & Strategic Deliverables

1. **Indigenous IP:** A locally developed sensor fusion and HUD rendering pipeline built entirely on open-source tools (Python, OpenCV, PyGame) without reliance on foreign proprietary software. This constitutes a standalone intellectual property asset.
2. **Integrated C2 Node:** A proof of concept demonstrating seamless fusion of 360-degree PiP situational awareness, live navigation minimap, digital compass, and IMU telemetry into a single soldier-worn system with sub-100ms response.
3. **Scalable Architecture:** A prototype that solves critical line-of-sight gaps while proving the networking and sensor infrastructure is ready for advanced integrations including thermal optronics and Head-Tracked Remote Weapon Stations.
4. **Defense-Grade HUD Interface:** A functional AR display styled after modern military HUD conventions — compass bar, tactical minimap, rearview PiP, and thermal scope window — demonstrating that Pakistani engineering students can produce defense-grade human-machine interfaces locally.

5. Estimated Component Cost

The following table covers the full helmet prototype:

FYP Cost Breakdown		
#	Component	Cost (PKR)
1	Raspberry Pi 5 (8GB)	59,000
2	ZT TFT SPI Module GC9A01	4,000
3	HUD Reflector Lens	400
4	Raspberry Pi Camera Module 3	17,000
5	Original IMX291 USB Camera Module	20,000
6	Digital Infrared Temp Sensor MLX90614	2,000
7	IMU Head Tracking	7,300
8	Helmet 3D Printed Prototype	TBA
9	Power Supply	500
10	Cabling and Wires	TBA
11	ESP32	1,500
12	Sony ECX339A OLED Panel	22,000
13	DLPDLCR2000EVM Pico Projector	35,000
14	Mini OLED Transparent Display 2x	30,000
15	Xreal X1 Pro (Back-up Option)	200,000
16	Operations & Logistics	70,000
17	Filament	30,000
18	Pulse Sensor Module	400
19	SpO2 Sensor	450
	TOTAL	**299,550**

6. Future Additions (If Time and Budget Allow)

- **UAV/Satellite Data Integration:** Live satellite or UAV imagery fed directly into the minimap display, providing the soldier with a real-time overhead view of surrounding terrain and contacts. For the prototype, a pre-loaded local map or Google Maps API feed will simulate this capability.
- **Inter-Helmet Networking:** Radio communication between multiple helmets in the same unit, displaying friendly positions as markers on each soldier's minimap. This removes the need for separate earpieces and radio devices, reducing equipment load.
- **Friendly Location Tags:** GPS-derived position tags that highlight allied unit locations in the AR HUD, enabling fast coordinated movement without voice communication.
- **Remote Weapon Station Video Feed:** If the turret platform is equipped with a camera, the live turret feed can be routed to the HUD as a secondary PiP window. This requires Wi-Fi (or 5G in a real deployment) rather than LoRa, as LoRa bandwidth is insufficient for video transmission.
- **Extended Thermal Integration:** Integration of FLIR Lepton or MLX90640 thermal core for full night-operation capability, including automatic hot-spot detection and threat highlighting within the thermal scope window.

Contact

Muhammad Daud — Team Lead, Electronics Engineering (Minor: Robotics & Autonomous Systems)

Email: khurramdaud333@gmail.com

Phone: 03410368528

Institution: Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Project Reference: FYP-27 /